



Adaptive mesh for shock capturing in CFD analysis of transonic airfoil RAE2822

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1. Problem statement

Although the cruise speed of common jet powered airliners is below the speed of sound, local regions of airflow around the wings may become supersonic. In these supersonic regions generation of shock waves occurs, and as the aircraft speed increases this phenomenon becomes stronger and produces noticeable changes in aerodynamic loads. These conditions correspond to transonic flight. Considering that nowadays modern commercial aircraft operate predominantly in this regime during cruise for most of their operational life, an inadequately designed airfoil geometry can lead to a significant decrease in aerodynamic efficiency and overall performance.

This report presents a steady two-dimensional CFD analysis performed with the open-source software SU2 [1].

Initially, CFD results will be compared with an experimental dataset obtained by Airbus [2] as a validation mean. This wind tunnel campaign was conducted on the well-known transonic airfoil *RAE2822*. The experiments were performed at the European Transonic Wind Tunnel pilot facility (pETW), a scaled environment that replicates the complex flow conditions encountered by airliner wings in transonic flight. The pETW facility offers a Mach number range from 0.15 to 1.3 and Reynolds numbers up to 230 million per meter, enabling accurate simulation of flight-relevant conditions.

After validation of the workflow through comparison with experimental data, a quantitative post-processing will be performed, through a comparison of aerodynamic coefficients trends with expected theoretical results, followed by a more qualitative reference to Schlieren images, a flow visualization technique commonly used to provide diagnostic information about the flow around the model.

2. Simulation setup

2.1. Geometrical domain

The geometry used for the simulations was directly taken from the *Airbus* work [2], which was adopted as the experimental reference. The geometry features a finite trailing edge to accommodate a pressure tap, instead of the sharp trailing edge of the original airfoil. The comparison between the actual experimental geometry and the ideal one (provided by *NASA* [3]) is presented in Figure 1, with a focus on the trailing edge.

As stated in Section 1, simulations were performed in a two-dimensional framework, and the computational domain was defined as an O-grid (Figure 2) having a diameter equal to onehundred times the airfoil chord. This domain size ensures that flow discontinuities and viscous effects can fully develop without artificial interference, while allowing the farfield boundaries to reliably represent undisturbed

free-stream conditions. In fact, from a theoretical point of view, it is common to take into account a distance of at least $(5 \div 10) * chord$ before and after the airfoil.

A specific treatment for a portion of the farfield was required for supersonic flight conditions though, treating it in fact as an outlet (see also 3.4).

The initial primary mesh was generated as an unstructured grid according to a Frontal-Delaunay 2D triangulation algorithm as offered by Gmsh [4] (Figure 2). This baseline discretization was subsequently refined through a mesh adaptation algorithm, enabling local resolution enhancement where needed, as described in 2.2.

2.2. Mesh Adaptation

Physical phenomena such as shock waves are characterized by strong anisotropic features, for which uniformly refined meshes, are generally inefficient in terms of both element size and orientation. For this reason, the *pyAMG* mesh adaptation software library [5] (already interfaced with SU2) was employed in the present study to improve the resolution of transonic flow features while maintaining a reasonable computational cost.

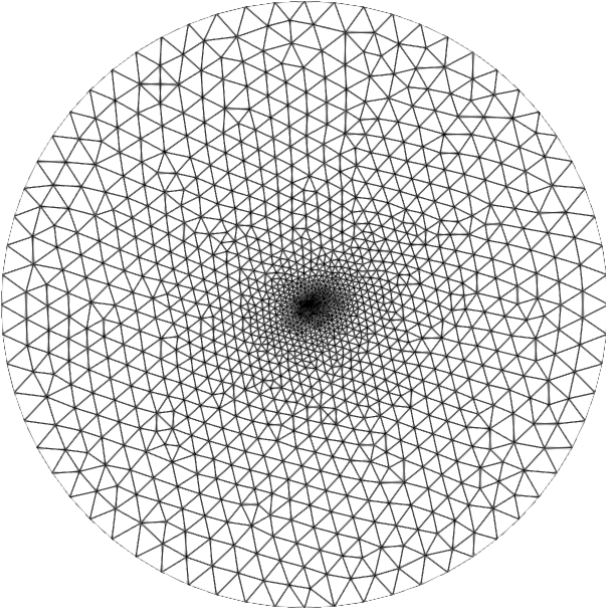


Figure 2: O-grid generated with Gmsh [4]

and robust solutions across the investigated range of Mach numbers and angles of attack.

The mesh adaptation procedure was carried out through a sequence of progressive outer adaptation stages, each consisting of a prescribed number of sub-iterations. At each stage, the adaptation criteria were gradually tightened, allowing the mesh resolution and anisotropy to increase in a controlled manner as the solution evolved.

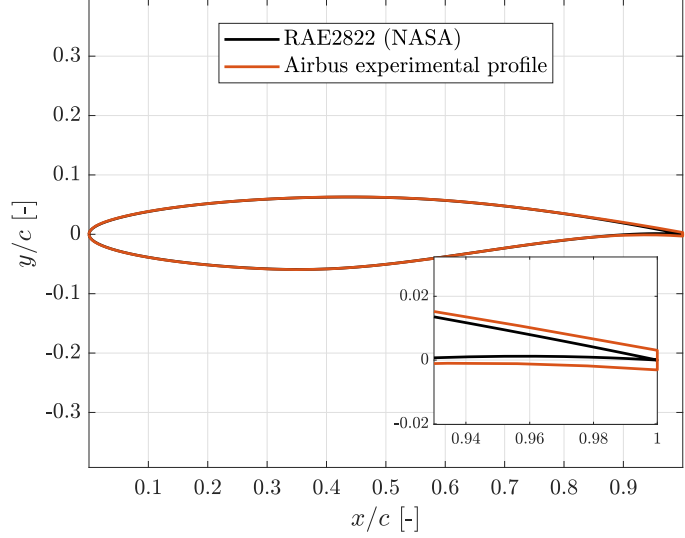


Figure 1: Geometry comparison

The adopted adaptation strategy is based on local gradients of selected flow quantities and exploits Hessian-based information to identify regions of high solution curvature. This approach enables primary triangular mesh elements (*simplices*) to be stretched and oriented along the principal directions of the solution curvature allowing grid nodes to be clustered preferentially in areas where sharp variations occur, such as shock waves, boundary-layer and wake, as well as shock–boundary-layer interaction regions, eventually enhancing the resolution of directional flow features, while keeping a coarser resolution elsewhere.

For the transonic *RAE2822* airfoil considered in this work, the Mach number was selected as the scalar field driving the adaptation process, as it proved to effectively highlight regions where phenomena of interest have place. Since the location of shock waves was not known a priori and varies with flow conditions, mesh adaptation proved particularly suitable for ensuring accurate

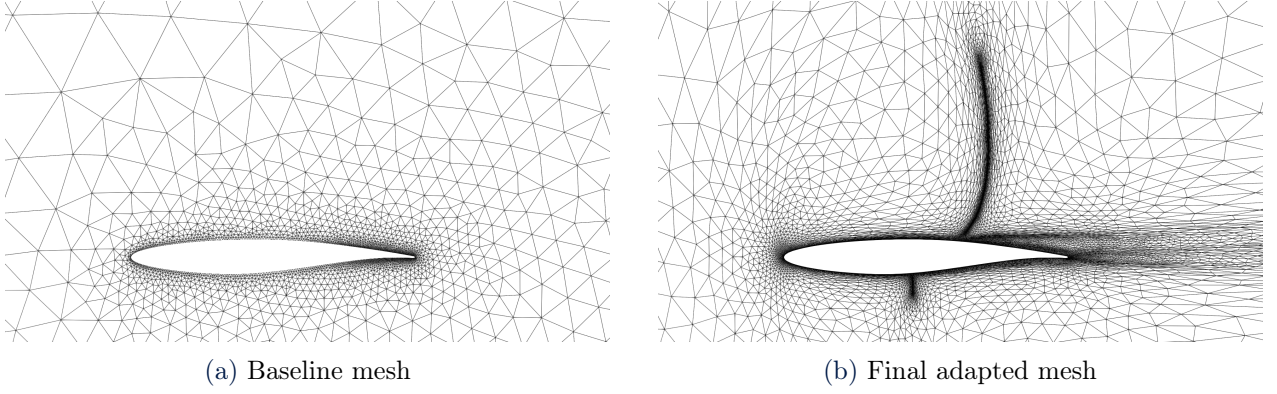


Figure 3: Overview of local resolution enhancement through mesh adaptation. The final adapted grid, though not a flow visualization per se, reflects the underlying flow features by locally increasing resolution in regions corresponding to shock waves and wake development ($AoA = 2.31$, $Ma = 0.8$)

Table 1: Mesh adaptation history for Airbus Case 450.

Iteration	Nodes
0	4328
1	11688
2	18537
3	26067
4	33550
5	35166
6	35954
7	37559
8	40185
9	43628
10	46171
11	49050
12	51390
13	53581
14	55423
15	56844
16	57805
17	58657
18	59239

During the initial three stages, only a limited number of sub-iterations¹ and flow iterations were performed, resulting in a moderate increase in flow features capturing capacity. This preliminary phase was intended to gradually tighten the adaptation criteria² and prepare the mesh for the final stage, in which a significantly larger number of sub-iterations was employed while keeping the adaptation parameters fixed. This allowed the mesh to repeatedly adapt to the evolving flow solution, enabling the shock-wave structures to progressively stabilize in space³. The final sequence of adaptation iterations was in fact intended to allow the shock position to converge toward a stable configuration.

It was observed that approximately fifteen adaptation sub-iterations in the final stage were sufficient to achieve a stable shock location and a consistent mesh distribution in the regions of highest gradients, i.e. a vanishing offset between mesh refinement and observable flow discontinuity locations. Also an increasingly demanding mesh complexity target was set for each outer-iteration, which ultimately led to meshes consisting of approximately 60000 nodes as shown in Table 1.

Achieving the same level of spatial resolution provided by mesh adaptation in the regions of interest using a single refined baseline mesh would have required an unmanageable number of nodes, thereby proving this technique to be indispensable.

The solution achieved after the last iteration of mesh adaptation was then considered representative of the physical flow field and used for the final analysis of aerodynamic coefficients and flow features. In Figure 3b is shown an example of how the baseline mesh was adapted.

3. Numerical methods

3.1. Governing equations

In order to reproduce conditions at which commercial and military aircraft operate in the transonic regime, the range of Mach numbers that will be analyzed in 4.3 is $0.5 - 1.2$, with a Reynolds number

¹This logic eventually led to the choice of a single sub-iteration for each of the first three outer-iterations

²Including lower boundaries for elements size and upper boundaries for their aspect ratio

³Since the mesh is regenerated at each adaptation step based on the gradients of the solution obtained on the previous grid, successive adaptations initially lead to small displacements of the shock location

equal to 6.5×10^6 . The compressible form of the RANS equations has been employed to account for the compressibility effects and to appreciate the interaction phenomena between the shocks and the boundary layer (e.g. λ -shock).

3.2. Convecting schemes

To effectively address the challenges of the transonic flight regime, the *Jameson-Schmidt-Turkel* (JST) numerical scheme was selected for the spatial discretization of convective fluxes. Given the use of mesh adaptation, the central JST scheme proved to be a more robust choice compared to upwind schemes based on MUSCL reconstruction with slope limiters⁴. JST combines a central discretization with adaptive artificial dissipation terms to prevent numerical instabilities. Namely, in regions of smooth flow, a fourth-order background dissipation is applied. However, in the presence of strong gradients, the sensor triggers a second-order dissipation term. This locally introduces sufficient numerical viscosity to suppress non-physical oscillations across the discontinuities. An adaptive *CFL* number, ranging between 1.0 and 10.0, was adopted as the best compromise between numerical stability and computational time. By ensuring a sharp and oscillation-free resolution of these supersonic regions, the numerical setup guarantees that the resulting noticeable changes in aerodynamic coefficients are physically representative and comparable with the pETW experimental dataset, thus fulfilling the validation purpose of this study.

3.3. Turbulence model

In the context of this study, the Spalart-Allmaras (SA) model was selected to close the RANS equations, driven by its proven robustness and widespread adoption in the field of external aerodynamics. Its formulation is specifically calibrated for wall-bounded flows, making it an industry standard for the analysis of transonic airfoils where boundary-layer separation may occur. The primary advantage motivating this choice lies in the model's exceptional balance between aerodynamic accuracy and computational efficiency. By solving a single transport equation for viscosity, SA allows for a precise resolution of critical flow features, such as the exact shock wave location and the complex λ -shock structure, without incurring the prohibitive execution times associated with multi-equation closures [6]. This efficiency ensures that the numerical simulation remains computationally affordable while delivering the high fidelity required for a reliable validation against the experimental dataset.

3.4. Compressible freestream and boundary conditions

For each simulation the following physical quantities were specified:

- Mach number;
- angle of attack;
- free stream temperature;
- Reynolds number;
- Reynolds length.

For supersonic flight simulations, a dedicated boundary treatment was required. Preliminary simulations exhibited spurious reflections of oblique shock waves at the farfield boundary, which required a large number of iterations to decay. To prevent this artificial behavior, the downstream portion of the farfield (i.e. the right semicircular boundary) was treated as an outlet by prescribing the free-stream static pressure. This approach effectively suppresses shock reflections while remaining consistent with the local flow physics. In fact, the imposed pressure is enforced only in subsonic regions, whereas in locally supersonic outflow regions the solution is entirely governed by outgoing characteristic waves, which starts from the airfoil profile. As a result, the imposed static pressure does not influence the solution where the flow is supersonic. Although this issue did not significantly affect the aerody-

⁴e.g. Roe scheme was observed to be more sensitive to mesh adaptation, often leading to convergence difficulties and, in several cases, divergence of the solution

dynamic loads, the adopted strategy effectively reduced the computational cost by limiting the number of iterations required for spurious oscillations at the external boundary to decay.

3.5. Convergence criteria

In this study, the numerical convergence of each simulation is monitored using the *RMS* of density residual. The solver will reach convergence if that sensor goes below 10^{-8} . As shown in figure 4a, the residuals consistently decrease after each mesh adaptation step, reaching successfully the target value. In addition, lift and drag coefficients exhibit a stable behavior after the initial transitional phase, which corresponds to the adaptation stages as well. These results ensure stability and reliability within the whole workflow and results.

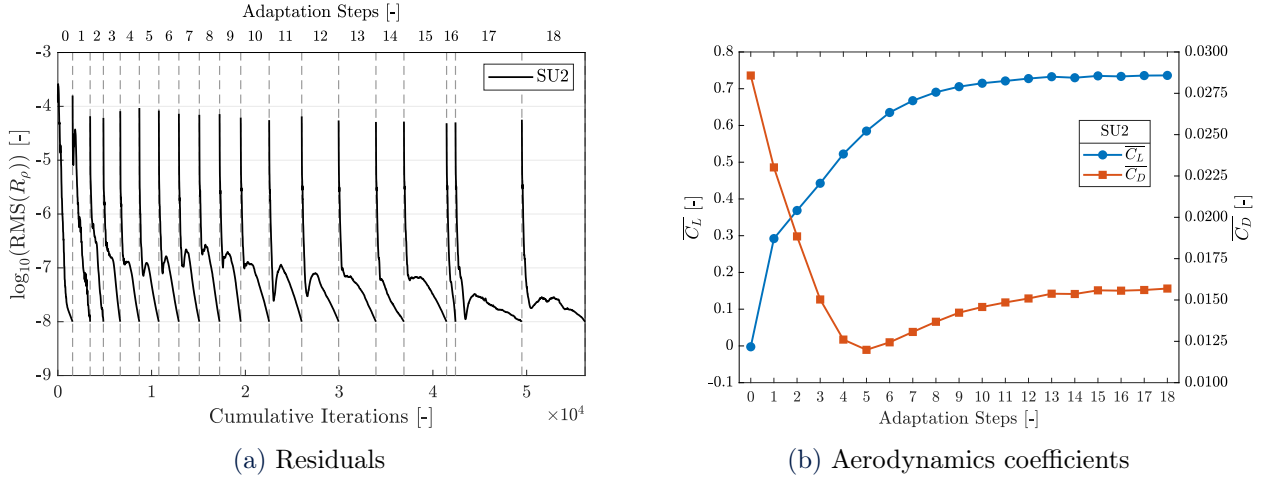


Figure 4: Convergence criteria

To ensure a fully converged and statistically steady solution, the simulation stopping criterion was not solely based on the residuals of the governing equations, but also driven by the stability of the integral aerodynamic quantities. Specifically, the percentage variations of the lift (C_l) and drag (C_d) coefficients between consecutive iterations were monitored. The reference C_l value at each step was computed as the arithmetic mean of the last 100 inner iterations; consequently, the percentage comparison was performed between the average of the current iteration and the average of the previous one. A threshold of 1.0% was established as the acceptance limit for these fluctuations. However, to rule out transient oscillations or premature convergence, the solver was not halted at the first instance this condition was met; instead, the simulation was terminated only after observing that the variation remained

Iteration	C_l	ΔC_l [%]	C_d	ΔC_d [%]
ite0	-0.002343	—	0.028576	—
ite1	0.292064	100.8023	0.023018	24.1438
ite2	0.368556	20.7543	0.018846	22.1412
ite3	0.442661	16.7409	0.015031	25.3819
...	...	...	...	...
ite13	0.732683	0.6940	0.015380	1.8760
ite14	0.729884	0.3835	0.015363	0.1149
ite15	0.734912	0.6842	0.015584	1.4205
ite16	0.733465	0.1973	0.015563	0.1321
ite17	0.735777	0.3143	0.015604	0.2604
ite18	0.736201	0.0575	0.015692	0.5611

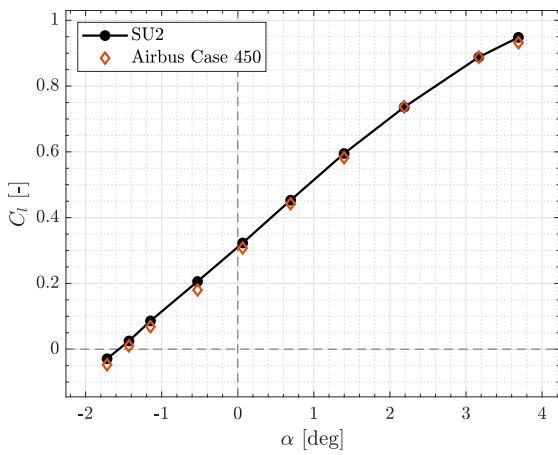
Table 2: Convergence of aerodynamic coefficients during mesh adaptation for Airbus Case 450

consistently below this threshold for a sustained number of iterations, confirming the achievement of a stable asymptotic state (an example is provided in Table 2). Particular attention was also devoted to achieving a stable spatial location of the shock-wave structures through successive final mesh adaptation steps, as discussed in 2.2.

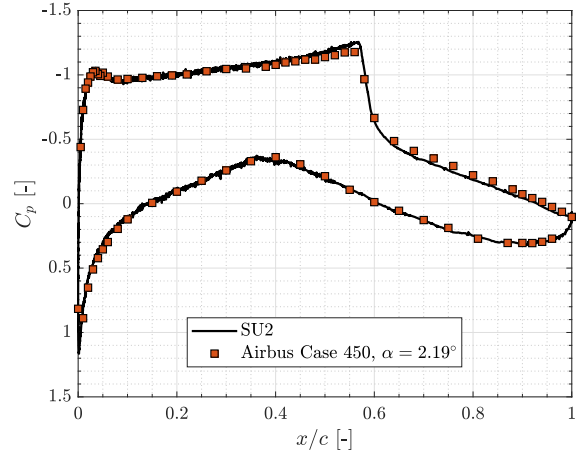
4. Results

4.1. Validation

The numerical workflow described above was rigorously validated against a comprehensive experimental dataset provided by Airbus, covering a wide range of transonic operating conditions. The validation campaign covered multiple combinations of Mach and Reynolds numbers, with full polar sweeps performed for each condition to assess the solver’s accuracy across different flow regimes. Among the available experimental test cases, Case 450 is selected and reported here as the most representative benchmark for validation purposes. For this specific test case a complete comparison of aerodynamic coefficients in the linear region is presented. Furthermore, to provide a deeper insight into the local flow physics and shock-capturing capabilities of the adopted numerical scheme, a detailed analysis of the pressure coefficient C_p distribution and shock wave position is reported for a specific angle of attack within this series. Figure 5a illustrates the linear region of the lift curve C_l vs α , demonstrating excellent agreement between the CFD predictions and the reference experimental data. Moreover, the pressure coefficient C_p distribution along the airfoil chord at an angle of attack of $\alpha = 2.19^\circ$, plotted in Figure 5b, confirms that the numerical solver accurately captures both the suction and pressure side distributions, as well as the precise location of the shock wave.



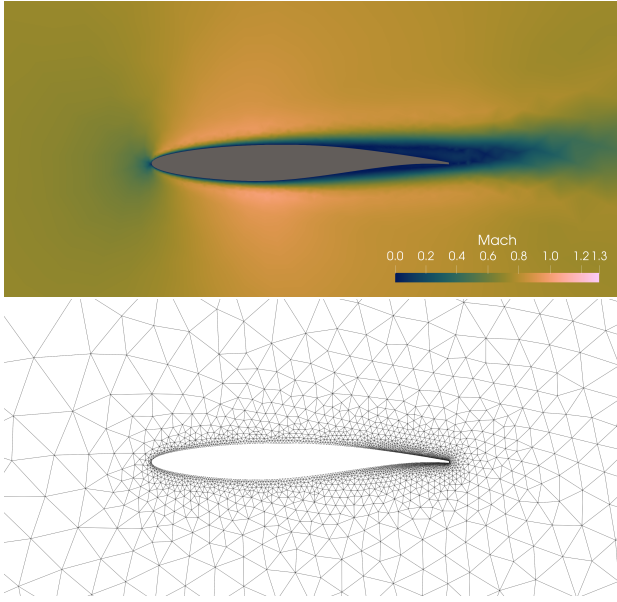
(a) Lift Coefficient vs. Angle of Attack



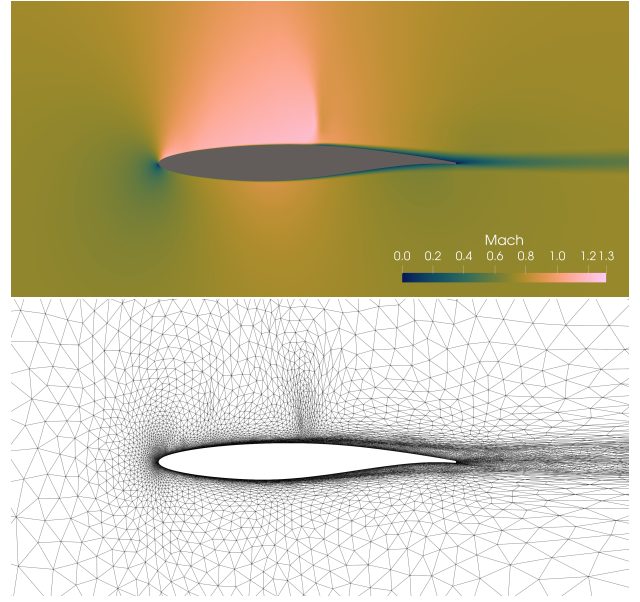
(b) Pressure coefficient distribution along the chord

Figure 5: Validation over *Airbus* [2] Case 450 ($M = 0.743$, $Re = 15.51 \times 10^6$)

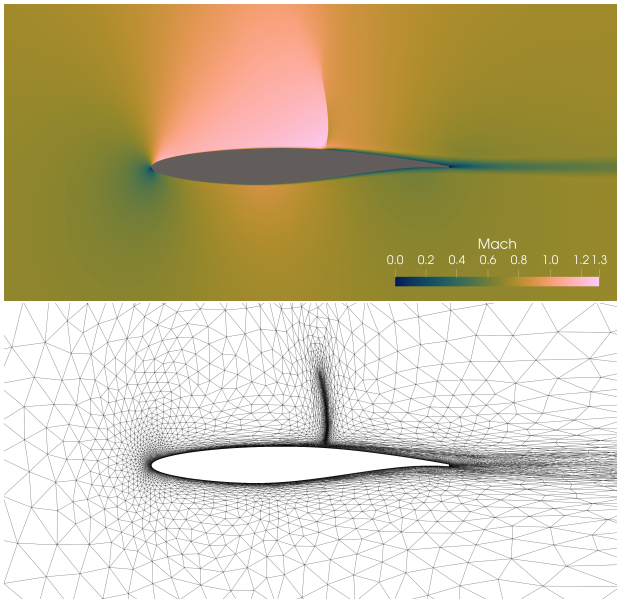
The close agreement between the available dataset and the CFD results, as shown in the plots, confirms the reliability of the numerical setup, allowing the prediction of global aerodynamic forces and shock position along the chord in the transonic regime. The mesh adaptation process leading to these results is illustrated in Figure 6, showing Mach contours and grid evolution at different intervals throughout the simulation.



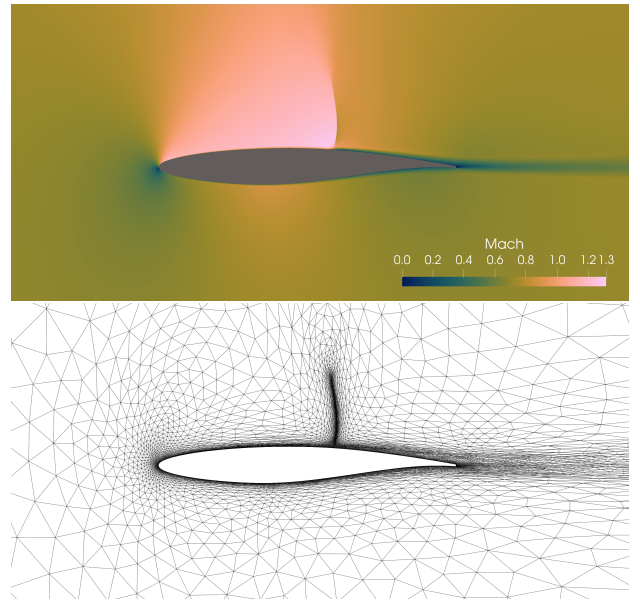
(a) 0 of 18 (initial solution)



(b) 6 of 18



(c) 12 of 18



(d) 18 of 18 (definitive)

Figure 6: Mach number contours and adaptive mesh for different steps (18 overall) of the adaptation process for simulation shown in Figure 5, *Airbus* [2] Case 450: $M = 0.743$, $Re = 15.51 \times 10^6$, $\alpha = 2.19$ (Colormap: ‘batlow’ [7])

4.2. Near-wall resolution (y^+) limitations with Adaptive Mesh

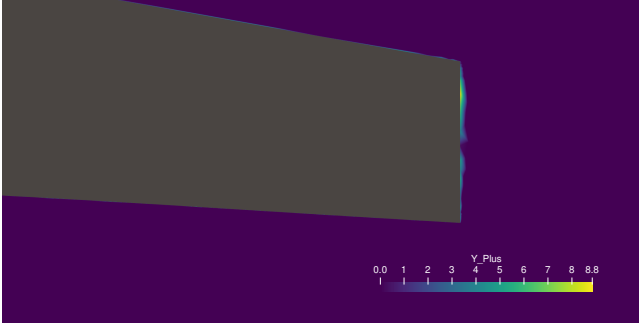


Figure 7: y^+ distribution over the trailing edge

The y^+ value was continuously monitored to ensure that it remained as close as possible to unity, since no wall functions were employed. However, due to the practical limitations of the mesh adaptation procedure, which does not allow for the targeted enforcement of a minimum cell size in selected regions, this requirement could not be strictly satisfied in some critical portions of the profile. In particular, variations were observed in low-velocity areas, such as near the trailing edge, where flow separation and the associated recirculation region are induced by shock-boundary-layer interaction.

A representative case of resulting y^+ parameter distribution in a critical region is shown in Figure 7.

4.3. Final study

As a final study, fixed angle of attack and Reynolds number were selected to highlight the effects of a varying Mach number on the aerodynamic performances of the *RAE2822* airfoil and on the corresponding flow topology in transonic regime. To do so, an analysis of aerodynamics coefficients trends and the corresponding Numerical Schlieren visualizations will be presented in this section.

An angle of attack of 2.31° and a Reynolds number of 6.5×10^6 were chosen to conduct this study in order to match an extra validation point (at Mach 0.729) with Case 6 of the *AGARD* work [8] and to get with that a direct comparison also on the drag coefficient, which was not present in the experimental data provided by *Airbus* [2]. These Mach and angle of attack values are the one corrected from wall interference effects and representing the equivalent free-air condition for *AGARD* Case 6 (the nominal experimental values were $M = 0.725$ and $\alpha = 2.92^\circ$), following the correction suggested by the *NASA* paper [9].

4.3.1 Aerodynamic performances

The resulting trends for C_l and C_d for varying Mach are shown in Figure 8a, where a lighter line color indicates the off-design regime: although the considered airfoil is specifically designed for transonic regime, within this study the Mach sweep has been extended up to Mach 1.2 to asses the effectiveness of the presented methodology and fully characterize the flow topology even in supersonic regime.

Over the C_l plot, a comparison with the linearized theory is presented. According to Prandtl-Glauert law the lift scales as

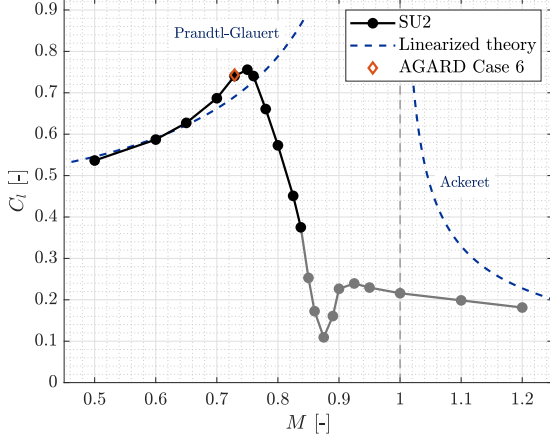
$$C_l = C_{l,M0} / \sqrt{1 - M^2}$$

within the subsonic regime. The starting point of the theoretical curve, $Cl_{l,M0}$ (lift coefficient at $M = 0$), has been obtained by correcting the lowest Mach cases provided by the *Airbus* work [2] to get the corresponding values at $M = 0$ and finally an interpolation over the angles of attack (valid within the linear aerodynamic region) has been made to match exactly the adopted value for this study.

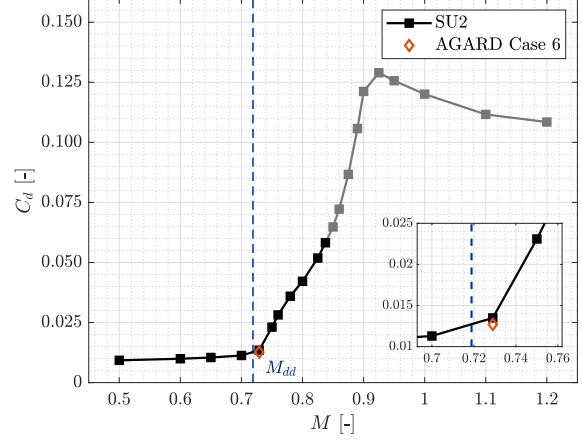
The Ackeret's theory holds on instead in the supersonic regime and it is formally derived for the flat plate. To account for airfoil thickness and camber effects, a simulation was performed at $M = 2$ and the resulting lift coefficient has been use to compute an effective angle of attack α_{eff} , which is subsequently applied to Ackeret's relation

$$C_l = 4\alpha_{eff} / \sqrt{M^2 - 1}$$

to draw the theoretical trend throughout the supersonic range.



(a) Lift Coefficient vs. Mach



(b) Drag Coefficient vs. Mach

Figure 8: Final study: Mach sweep ($\alpha = 2.31^\circ$, $Re = 6.5 \times 10^6$)

Moreover, in figure 8b a vertical dashed blue line representing the *Drag Divergence Mach* (M_{dd}) computed according to the "slope definition" ($\frac{dC_d}{dM} = 0.1$) was reported (the obtained value from the simulations results is $M_{dd} = 0.7189$).

As known, the linearized theory features a singularity approaching sonic Mach, leading CFD to be one of the few possible options to characterize the flow in transonic regime, together with classical visualization techniques, such as Schlieren imaging.

The lift coefficient initially follows the Prandtl-Glauert compressibility scaling and then starts to show the typical overshoot around $M \approx 0.7$. Beyond the peak, the sharp drop in C_l (shock stall) is driven by the strong interaction between the shock wave and the boundary layer, leading to extensive flow separation downstream [10].

The drag coefficient is dominated by viscous friction in the subcritical regime, showing negligible variation until the *Drag Divergence Mach* number is reached. The subsequent exponential rise in C_d has to be attributed to wave drag. The resulting curve highlights also the typical drag peak which has to be overcome to fly in supersonic regime.

Some of the most relevant points of these trends are shown by means of Numerical Schlieren images in Figure 9, including the *bow shock* (detached shock in front of the airfoil) visualization for $M = 1.2$ in Figure 9f.

The comparison with *AGARD Case 6* shows excellent agreement for C_l (0.7404 vs 0.743, -0.35% difference), while C_d (0.0135 vs 0.0127) is overestimated by $+6.24\%$ ($+7.9$ drag counts). This discrepancy, highlighted by the zooming box in Figure 8b, falls precisely within the expected bounds and it has to be attributed mainly to the *fully turbulent* characteristic of the SA model described in Section 3.3, which generates higher skin friction near the leading edge compared to the experiment where transition was fixed at 3% chord.

4.3.2 Numerical Schlieren

Schlieren imaging is an optical method sensitive to the first derivative of fluid density, based on the refraction of light rays passing through a transparent medium. According to the *Gladstone-Dale* relation, the refractive index n of a gas is linearly coupled to its density:

$$n = 1 + K_{GD} \cdot \rho$$

where K_{GD} is the *Gladstone-Dale* constant. Consequently, light rays traversing the flow are deflected by density gradients ($\nabla\rho$), creating the shadow patterns that make compressibility effects visible.

To replicate this diagnostic numerically, the magnitude of the density gradient vector ($|\nabla\rho|$) is computed over the computational grid. However, since the gradient values can vary by orders of magnitude between the shock wave and the smoother wake regions, a linear visualization would fail to capture

details. To solve this, a non-linear processing step is applied using an exponential transfer function:

$$I \propto \exp(-k \cdot |\nabla \rho|)$$

Here, the scaling factor k acts as a "virtual sensitivity" parameter, replacing the optical tuning of the experimental setup and it was determined through an iterative process to reach the optimal appearance. The result obtained is shown in Figure 9, where a few key Mach values among the whole range of the final study reported in Figure 8 are considered.

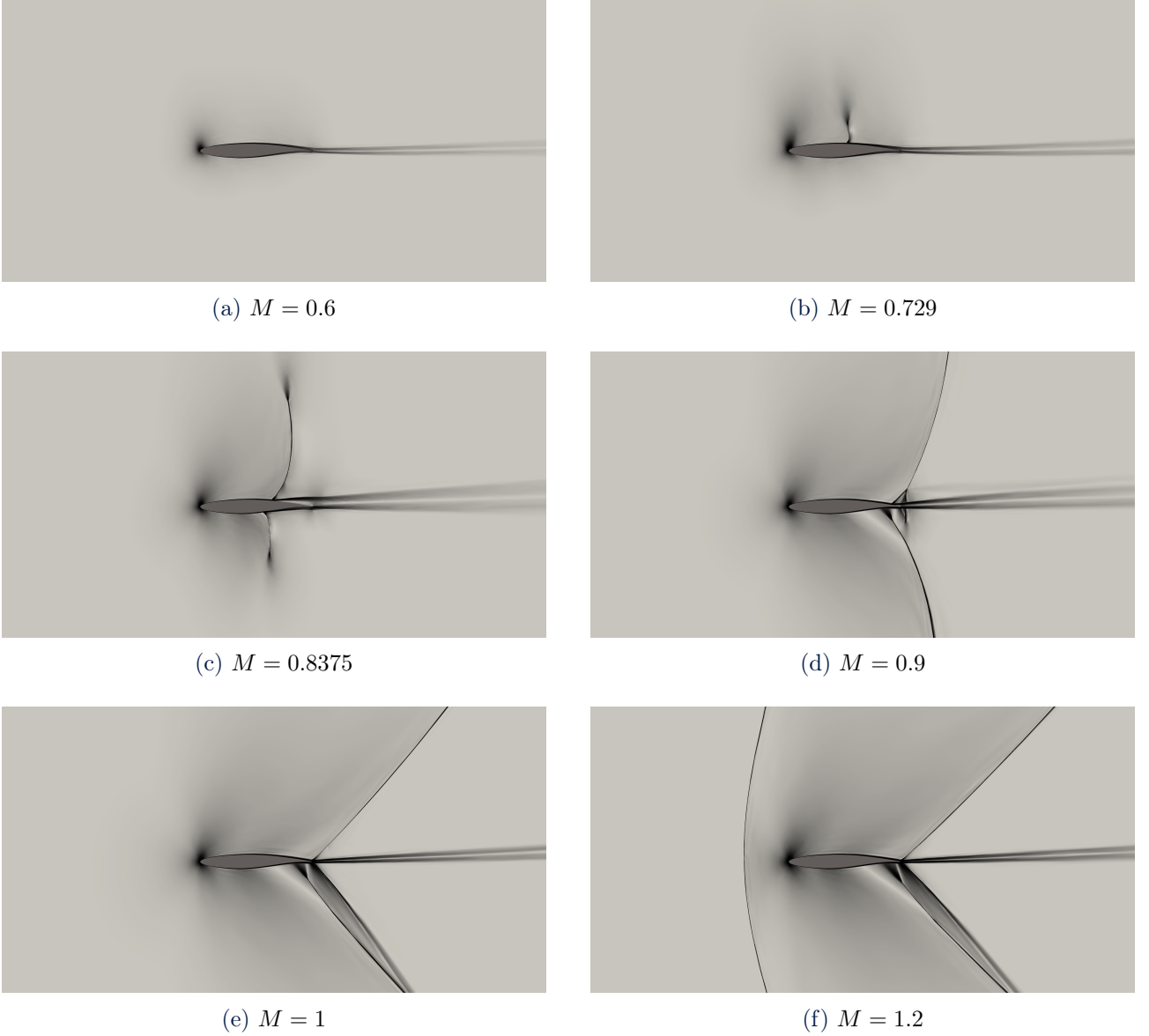


Figure 9: Numerical Schlieren visualization for relevant stations of the Mach sweep shown in Figure 8 ($\alpha = 2.31^\circ, Re = 6.5 \times 10^6$)

5. Conclusions

In this work, a CFD methodology for the numerical characterization of the transonic flow around the *RAE2822* airfoil has been developed and assessed through detailed comparison with experimental data coming from a wind tunnel campaign made by Airbus. An accurate combination of adaptive mesh refinement, tailored numerical methods and boundary conditions proved to be efficient in capturing the physics aimed for this project. Moreover, the adopted strategy worked properly to reduce spurious numerical oscillations, ensuring stable convergence and stability within simulations. The computed

aerodynamic coefficients showed very good agreement with the available experimental measurements, confirming the suitability of the proposed approach for the analysis of transonic flows.

Despite the relatively low computational power, simulations showed satisfactory results with optimized resources. It was possible to reproduce correctly the flow physics in steady case, as shown in comparisons with theoretical (Prandtl-Glauert and Ackeret) and qualitative results (Numerical Schlieren).

Overall, the results demonstrate that the proposed CFD methodology represents a reliable workflow inherent in transonic regime, making it useful for preliminary design of an airfoil similar to the *RAE2822* in industrial applicability.

In conclusion, a future development of the presented study could be, possibly with higher computational power, the extension to three-dimensional wing configuration or an unsteady flow simulation, analyzing the transition and acceleration from subsonic to supersonic regime.

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